

Visualizing Internet Traffic Using TCP Traceroute

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Jack Crouse

Department of Computer Science
Furman University
jcrouse2@furman.edu

Abstract

While accessing content through the internet might seem simple, there is a lot more going on than meets the eye. Packets of information can travel thousands of miles, making many stops along the way before returning to the user. But where is this information actually going?

The goal of this project was to answer this question by providing a meaningful geographic visualization of how internet traffic travels from point A to point B. This was done by implementing traceroute and linking it to a web application. The website (<https://jacrouse.github.io/>) performs a traceroute on whatever hostname is specified by the user. In addition to the web server, several hosts were automatically scanned over time to provide statistics and insight into how paths change.

The portion of the project pertaining to the web application was successful. It quickly provides the results of the traceroute and plots each hop's geographic location onto a map, tracing a path. The data collection was also a success. Dozens of paths were successfully traced and recorded, which were then used to create statistics and geographic maps. While satisfactory, the results could be further refined by collecting more scans on a wider array of hosts. Investigation into different techniques of performing traceroutes could also prove beneficial.

1 Introduction

To support over 5.3 billion global internet users there exists an extremely complicated system of networks across the globe [1]. These networks orchestrate the delivery of information through a complicated system of routers that maintain a balance between speed, cost, and network performance. During this process information can travel using many mechanisms, such as radio waves, undersea fiber optic cables, or even satellites in low earth orbit. TCP (transmission control protocol) is the one of the protocols used to exchange information between hosts on the internet. It is a robust protocol that focuses on establishing a viable connection before exchanging data. This project attempted to capture the geographic path taken by internet connections using TCP traceroute. This was done in an effort to shed light on how complicated the "behind the scenes" of using the internet really is.

The first part of this project was focused on a small website that allows users to input any host. After a host is selected, a traceroute is performed in real time and the results are displayed to the user. The results consist of a map where each hop is marked and a line is drawn through them chronologically. The actual technical results are also printed below the map, and consist of each hops time to live (TTL), the status of the traceroute (origin, intermediate hop, or destination reached), the source IP address, and the geographic coordinates. Time to live is a number attached to packets that indicate to each router how much longer the packet can be forwarded before it should be destroyed.

The IP address is an identifier made up of an octet of numbers (ex. 192.168.1.1) that is used to identify and locate devices.

The second part of the project was an attempt to collect data by performing routine traceroutes on a collection of carefully selected hosts. The data collected consisted of detailed reports of each traceroute, such as the overall time it took to perform and the path taken. The results were used to create several visualizations depicting statistics like average latency by host and the variance among the paths taken. This was done in order to try and establish meaningful patterns about the paths taken.

There is a lot of research about analyzing the paths taken by internet traffic but the majority of it uses ICMP traceroute. ICMP, or internet control message protocol, is another common protocol used on the internet. It is used to transmit simple information such as a host's status. During the first stages of this project, ICMP traceroute was used, but with very limited success. The explicit reason for this is unknown but intermediary routers will sometimes drop ICMP packets because they can be interpreted as non-essential. This can result in limited information about what nodes are visited being returned to the tester, which is likely what happened in this case. For this reason it was decided that TCP traceroute would be used, because TCP packets are less likely to be dropped on account of them being used in the majority of internet traffic. There was a marked improvement in the detection of each node when the underlying method was switched from ICMP to TCP.

2 Related Work

Traceroute is an extremely popular network-diagnostics tool. Despite its old age it comes preinstalled on Windows, OSX, and many Linux distributions. It allows network-connected devices to discover the path taken to a host step-by-step. This is useful for a variety of reasons, particularly when it comes to debugging network issues, mapping network topology, and locating possible performance problems. The most popular purpose it serves in academia is path-tracing, with the network often represented as a graph and the hosts as nodes. There are a huge quantity of papers that use traceroute in ordinary and sometimes unorthodox ways, but it is not always reliable and has been known to have unpredictable behavior. Paths from node to node are often fickle, and might change due to external circumstances like load-balancing. It is also possible for hosts to ignore the protocol entirely, which can create gaps in observed paths. Because this project relies on traceroute to map the paths taken to hosts, the literature review will focus on examples of traceroute being used in academia for similar forms of research. In doing so it will attempt to shed light on the current role traceroute has in academic literature, as well as the advantages and limitations it provides in research.

The first piece of public information on traceroute is the original manual page from 1988. The page was written by Van Jacobson and contains useful information about how to use the tool and how it works. For the sake of credit it's worth noting that Van Jacobson said he didn't come up with the idea, his friend Steve Deering did. He summarizes the project as an "attempt to trace the route an ip [*sic*] packet would follow to some internet host." He goes on to say how he discovered why he had been having trouble sending mail to MIT, which is likely one of the first times this program had ever been ran. Among these examples he displays the paths taken each step of the way to various servers he performed traceroute on. This very early use of In addition, it highlights the first and most important use-case of the tool: debugging networks by observing the flow of traffic. [2] While this source is not directly academic it is included because its the the first and technically the most important piece of material on traceroute ever released. Its worth pointing out how the usefulness of this tool has not dwindled as time moved on. It is still used for the same purposes now as it was back in the day. Because traceroute is a tool and not an area of study in and of itself, from here on out the sources discussed will gravitate towards research that heavily leans on using traceroute.

This area of research almost exclusively uses traceroute or one of its derivatives. The paper "The End-to-End Effects of Internet Path Selection" is one such article. It focuses on determining how effective internet routing is for the end user. The authors were concerned with what specific circumstances introduced entropy into the selection of paths. To do this, they used "five datasets containing large numbers of Internet path measurements taken between geographically diverse hosts." The paths and some information were recorded using traceroute. Various metrics

were included in the conclusion of their research, such as statistics on latency, packet loss and the performance impact of the time of day. They had a heavy focus on the performance of alternative paths. The difference between this project and theirs is that there is more of a focus on the geographic locations rather than the types of networks that packets travel through/what causes change in paths. [3]

While the majority of research on analyzing internet paths came from using traceroute, some did not. DSL, or a digital subscriber line, is technology that allows the transmission of data over telephone lines. An ISP, or internet service provider, is an organization that provides end users with internet access. For example, "On Dominant Characteristics of Residential Broadband Internet Traffic" used a data set obtained directly from "a major European ISP" that contained "passive packet-level monitoring of more than 20,000 residential DSL lines." Unfortunately for this project, this is not really an option. There were various datasets for sale at the time of this research, but they were very expensive. Because they had access to this data, they focused their study primarily on what type of traffic existed and what the underlying characteristics of alternate paths were. [4]

The paper "Measuring ISP Topologies with Rocketfuel" is a highly cited paper that contains the most similar research to this project. They also used traceroute to map their paths. Their goal was to create topological maps of internet service provider networks. In addition, they were concerned with how to perform a large number of traceroutes without causing excessive burden. This resembles the goal and concerns of my project very closely. The differences are the same as the last paper, particularly that the focus of this project is on geographical location rather than just finding network paths. They also highlighted another phenomenon. As soon as one is finished making a map there is a high likelihood that the topology would have shifted from how it originally was. Another interesting problem they ran into was how to resolve an IP address to its alias. This research accomplished this through the use of a third party API service. [5]

Another paper called "The Flattening Internet Topology: Natural Evolution, Unsightly Barnacles or Contrived Collapse?", used traceroute to find trends that large tech companies were moving towards that could become potentially harmful to users. While using traceroute, they recorded how involved ISP nodes were on the way to their targets. They knew that multiple popular websites fell underneath the ownership of Google and were hosted in Google's network. The trend they noticed was that companies like Google were geographically placing their servers as close as possible to users to provide faster speeds. This change could potentially harm tier 1 ISP's by rendering them less necessary. Users would require the services of ISP's less if Google is just one hop away, and so the topology of the internet would become flatter. This influenced the decision to try and see if there are changes to the amount of hops it takes to get to a host over time. In other words, you could try to estimate if the paths are becoming shorter or longer by observing how many stops are taken along the way. [6]

There is a sizable area of research that uses traceroute in unorthodox ways. There were some interesting writings, such as "Reverse Traceroute" and a section of a journal called "Measure the Earth's Radius and the Speed of Light with Simple and Inexpensive Computer-Based Experiments." The former aimed to resolve potential issues in using ordinary traceroute by making their own version dubbed "reverse traceroute." Essentially this would perform traceroute from the destination back to the source, which is the opposite direction as a typical traceroute. They found their tool to be statistically more effective. The latter of the two articles was about using traceroute and ping to measure the diameter of the earth as well as the speed of light. The theory behind this is recording how long it takes to reach a host through cables. [7] [8]

Several articles have been written detailing drawbacks and problems with traceroute when it comes to accuracy and reliability. For example, "Don't trust Traceroute (Completely)." It extensively detailed issues like ghosting routing changes and overestimating load balanced paths. After repeated traceroutes they found that a shocking 32.1% of the paths never actually changed despite traceroute suggesting otherwise. For this reason TCP traceroute was what this project ended up using, as it proved to be more accurate. [9]

The paper "Avoiding Traceroute Anomalies with Paris Traceroute" details how to get around the limitations of ordinary traceroute. Paris traceroute is designed specifically for routers that use load balancing, due to load balancing typically throwing off normal results. It offers features like telling the user whether there is per-packet based or per-flow based load balancing. It also included a lot of information about possible obstacles that routes can run into, such as loops and cycles. [10]

3 Data

Data collection was performed by writing a python program that scanned several hosts routinely. This program was run from the same server that hosted the website, which was located in New York City, NY. The server was rented from DigitalOcean. The hosts were selected due to their geographic location. This was because many international websites are cached or hosted in the United States. To learn more about long distance traffic, hosts that were cached in the US were deliberately avoided.

Location	Hostname	Distance From Server (km)
Kansas City, MO, United States Alt (www)	www.redfishportorford.com	1,758.92
Los Angeles, CA, United States	redfishportorford.com	3,935.74
Moscow, Russia	mail.ru	7,510.31
Tehran, Iran	sbu.ac.ir	9,856.23
Ulaanbatar, Mongolia	news.mn	10,158.24
Tokyo, Japan	webtan.impress.co.jp	10,848.68
Shanghai, China	ftp.sjtu.edu.cn	11,859.49

Figure 1: A chart containing information about the hosts selected for data collection

A total of 91 traceroutes were performed on these 7 hosts, with each scan being done every 2 hours. The scans were recorded into a csv file using pandas. The data recorded included the destination of the scan, the time to live, the status at each stop, the source IP address of the stop, the geographic coordinates, and the overall roundtrip latency. Excel and Tableau were used for calculations and creating the visualizations. Some of the graphs created using the entire data set are pictured in the results section.

Destination	TTL	Status	Source	Coords	Latency
webtan.impress.co.jp	0	Origin	167.99.0.220	[40.7127837 -74.0059413]	-1
webtan.impress.co.jp	1	Intermediate	138.197.251.86	[40.7127837, -74.0059413]	-1
webtan.impress.co.jp	2	Intermediate	138.197.251.118	[40.7127837, -74.0059413]	-1
webtan.impress.co.jp	3	Intermediate	138.197.244.7	[40.7127837, -74.0059413]	-1
webtan.impress.co.jp	4	Intermediate	10.70.134.111	['NoLat', 'NoLng']	-1
webtan.impress.co.jp	5	Intermediate	129.250.3.8	[40.735657, -74.1723667]	-1
webtan.impress.co.jp	6	Intermediate	157.238.179.69	[41.8781136, -87.6297982]	-1
webtan.impress.co.jp	7	Intermediate	129.250.2.166	[40.7281575, -74.0776417]	-1
webtan.impress.co.jp	8	Intermediate	129.250.5.17	[40.7281575, -74.0776417]	-1
webtan.impress.co.jp	9	Intermediate	129.250.6.131	[35.6940027, 139.7535951]	-1
webtan.impress.co.jp	10	Intermediate	61.213.161.50	[35.7090259, 139.7319925]	-1
webtan.impress.co.jp	11	Reached	35.74.58.61	[35.7090259, 139.7319925]	201.2569904

Figure 2: Example entry from the data set

4 Methods

The most important part of this project was the choice of which type of traceroute to use to maximize accuracy. At first, ICMP traceroute was used, which is how traceroute is classically performed. Experimentation proved that this method was not satisfactory. There were large gaps present in traced paths. This is likely due to routers dropping ICMP packets, which can sometimes be interpreted as unnecessary or spam. For this reason, TCP traceroute was used instead. There were significantly less issues with using TCP traceroute, which is likely due to TCP being an essential part of how most services work. In theory, the importance of TCP connections means that routers are less likely to drop them as opposed to ICMP. The traceroute used was implemented in Python, using Scapy, which is a packet crafting library.

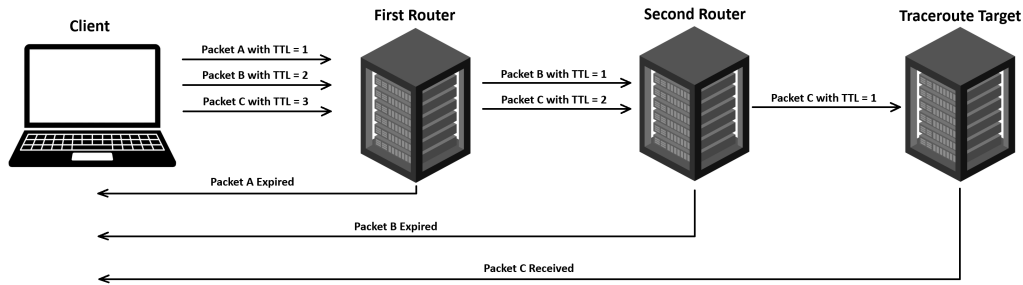


Figure 3: TCP traceroute diagram

Because its not possible to perform a traceroute from a clients browser, the website concurrently ran a flask server that performed the traceroute for the client. When a client inputs a host and submits, the client's browser contacts the flask server using an HTTP POST request. This post request contains the target website's information. The flask server then records the clients IP address and performs a traceroute on the requested website. A 3rd party API is used to resolve both the clients IP address and the IP addresses from the traceroute results to a geographic location in the form of coordinates. Once this process is complete, the results are sent back to the client, where they are then plugged into Google's Google Maps API and a path is drawn.

The data collection was much more simple. It was a python program that scans a list of hosts and stored the results in a csv file. This was done using the Pandas library in Python. After its finished, it waits 2 hours and then repeats the process. This is done until the program is manually stopped. The server that ran both the website (apache2), the flask server, and the data collection, used Debian Linux. The processes were all ran concurrently using screen and systemctl. The specifications of the server are listed below.

```
root@debian-s-1vcpu-1gb-nyc1-01
OS: Debian GNU/Linux 12 (bookworm) x86_64
Host: Droplet 20171212
Kernel: 6.1.0-18-amd64
Uptime: 16 days, 3 hours, 42 mins
Packages: 712 (dpkg)
Shell: bash 5.2.15
Resolution: 1024x768
Terminal: /dev/pts/0
CPU: D0-Regular (1) @ 2.294GHz
GPU: 00:02.0 Red Hat, Inc. Virtio 1.0 GPU
Memory: 265MiB / 960MiB
```

Figure 4: Results of neofetch

It is important to note that every type of traceroute, including TCP traceroute, is not always perfectly accurate. Although TCP traceroute performed much better than the classic ICMP traceroute experimentally, it has its own problems. Because TCP traceroute relies on multiple packets being sent and received, it's possible that the path can change in the middle of a traceroute, which would provide inaccurate results that appear normal. Another issue was that some of the traced paths contained repetition, as if the same router was visited several times in a row. Because of the nature of using traceroute, unusual phenomena are often unexplainable. The transparency of the path taken is generally at the mercy of how the routers in between the source and the destination are configured. While hosts cooperated more with TCP, it's still possible that sent packets can be dropped along the way, leading to incomplete paths.

This project also used a third party API to resolve the geographic coordinates of hosts. This means that the geographic data obtained could possibly be inaccurate, depending on where the API was getting its data from. At the beginning of this project I tried to use the "whois" command which pulls from the Internet Corporation for Assigned Names and Numbers (ICANN) database. This was difficult to parse effectively, and sometimes resulted in a complete lack of geographic information, so it was scrapped. I also experimented with getting datasets that have IP address range geographic information. Every dataset I could find was expensive, and even if you had one it could become outdated or inaccurate after a short period of time. A significant amount of time spent on this project was dedicated to making it work as accurately as possible, but even with more time and resources these limitations would be extremely difficult to solve. The same pattern of problems that I ran into were present in several of the academic papers I reviewed in preparation for this project.

5 Results

Establishing meaningful patterns using the results of TCP traceroute is a difficult task. Factors such as the configuration of routers, load balancing, and redundant storage of content mean that the path taken at any given time is highly unpredictable. Performing research using more data, different techniques, or tools like Paris Traceroute and MTR (My Traceroute) would be useful in learning more about internet topology.

The website performed exactly as described. If given a valid host, it quickly delivers the results of traceroute and plots them accurately on the map. Some interesting observations were how quickly paths can change. Repeatedly tracing the same host can result in the exact same or very different paths, depending on the host. This could be happening for a multitude of reasons. One is load balancing, which is when DNS (domain name servers) servers or routers manipulate the flow of traffic so that the workload is more evenly distributed. This helps prevent congestion that could occur due to networks becoming overwhelmed. Another interesting phenomenon was that foreign high-traffic hosts will often times appear to exist conveniently closeby in the United States, despite this not being where their main servers reside. This is likely due to systemic caching of certain hosts through services like content delivery networks (CDNs), which is done to increase speed and availability for users. This phenomenon appears to be less likely to occur with smaller, low-traffic websites.

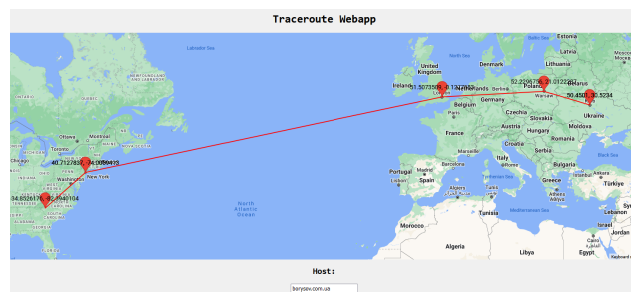


Figure 5: Results of a trace on a website located in Kyiv, Ukraine

The results from the data collection were also interesting. Traffic sometimes takes seemingly nonsensical paths. The data collected recorded multiple instances of traffic bouncing around the United States before leaving the continent. Interestingly, the inclusion or exclusion of “www.” before a website can result in a completely different path/destination. This could be due to DNS servers providing different information for the two hostnames, despite them both providing the same website. It could also be CDNs caching the same website in different locations. Below is a visualization of many of the paths taken. The hosts are color coded, meaning the same color path is the same host. Some of the hosts in this graphic were removed to make it easier to read.

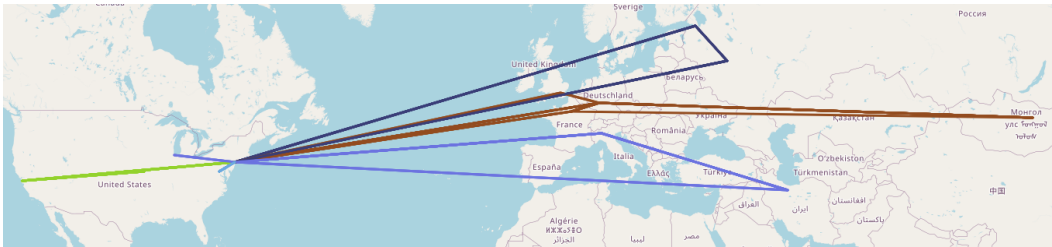


Figure 6: The many different paths traced over unique hosts

Below are bar plots of some of the statistics gathered from the paths collected. The values in the "Consistency of Devices Visited by Country" plot were calculated by taking the ratio of devices that were present in every scan to devices that were not. The reasons the data included two versions of the same website, one with "www." and one without, was to see if it makes any meaningful difference in the path taken or its characteristics. The average/variance of the number of stops taken and the consistency of the paths taken were different between the two hostnames. Another interesting observation is that the host in Mongolia (the third farthest away at over 10,000km) was extremely consistent in the nodes visited. Before data collection I suspected that the greater the distance traveled, the less consistent the nodes there would be. My hypothesis was that if the packets had to travel farther there was a greater chance of them taking different routes due to the greater number of potential nodes, yet the data suggests otherwise.



Figure 7: Bar plots showing various information about each host

To visually determine the frequency of nodes being visited, I made a geographic map of the nodes visited and colored them in accordance with how frequently they showed up in the data. Unsurprisingly, there were a huge number of hits in NYC where the server is located. This was primarily for three reasons, the first being the servers IP itself being resolved each scan. Secondly, there were instances where the destination would exist in NYC, which could be due to load balancing and content caching. Thirdly, some of the traced paths would stop in NYC/New Jersey on their way out towards their host, which is most definitely due to some sort of load balancing, especially because it didn't happen all the time.

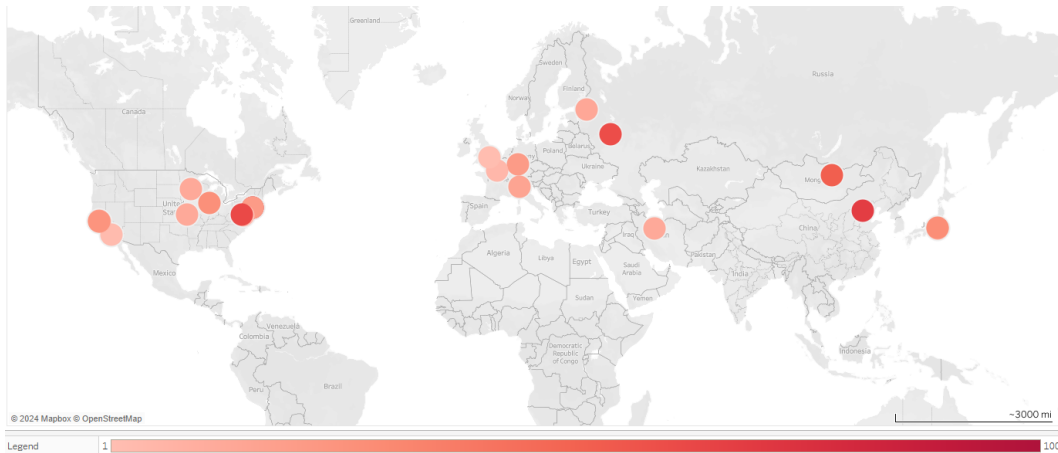


Figure 8: Frequency map of locations visited during scanning

City	Frequency
NYC/Jersey City	427
Washington, D.C.	1
Ashburn, VA	62
Chicago, IL	27
Minneapolis, MN	13
Kansas City, MO	13
Los Angeles, CA	3
San Francisco, CA	50
Tokyo, Japan	40
Beijing, China	68
Ulaanbaatar, Mongolia	52
Tehran, Iran	13
Moscow, Russia	60
St. Petersburg	14
Milan, Italy	16
Frankfurt, Germany	21
Paris, France	5
London, UK	2

Above is a table of all of the locations visited, with their total count. I had some confusion as to why Ashburn, VA made this list. I had never heard of Ashburn before, and it turns out it's a pretty small city. Upon further investigation I learned that Ashburn is a very popular location when it comes to data centers. The area is apparently extremely important for local and international internet traffic. [11] This raised an interesting question: if enough servers performed data collection in the same manner as this project, could you locate important data centers? An interesting experiment could be to quantify how important data centers are by measuring what proportion of traced paths include them. This would have to be done from many different locations to avoid any local bias in paths taken.

6 Future Work

If this type of experiment could be repeated for a longer duration of time, it would be easier find patterns in the traced paths. Although a large amount of data was used, it was collected in a short period of time. It would be interesting to see if there are more significant changes over a period of weeks or months. In addition to a greater volume of data, it would be worth investigating if other traceroute techniques are superior to TCP traceroute in performance. More complicated and robust methods of traceroute include Paris Traceroute and MTR (My Traceroute).

As stated at the end of the results section, there are several possible projects that could make use of the same, or similar, methodologies. Below are some possibilities:

- Using multiple servers, you could collect data on which ISP's or data centers most frequently show up in traced paths.
- A study comparing the results of using ICMP, UDP, and TCP traceroute.
- Creating homemade versions of traceroute by drawing from existing variations.
- Experimenting with unorthodox uses of traceroute, like measuring distance.
- Attempting to detect and visualize when undersea cables or satellites are used.
- Creating a dedicated application to visualize traceroute, rather than a web application.
- Automatically detecting network outages by monitoring traced paths.
- Creating a large dataset filled with traced paths.
- Constructing a graph of internet hosts using the results of traceroute.

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